

EZRadio and EZRadioPRO Wireless Solutions

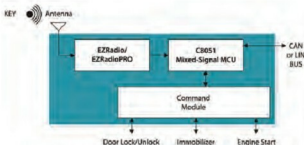
Complete Family of Transmitters, Transceivers and Receivers

Silicon Labs' family of EZRadio and EZRadioPRO single-chip transceivers, receivers, and transmitters are highly integrated, low power, low cost solutions designed to support a wide range of wireless applications. The EZRadioPRO family features industry leading performance and an extensive set of advanced features to reduce overall system cost and complexity. The EZRadio family offers a complete solution for applications where cost and space are at a premium while still supporting such advanced FSK features as frequency hopping. EZRadio and EZRadioPRO support proprietary and standards-based point-to-point, star, and mesh networks.



Example Solution: Automotive Remote Keyless Entry

Silicon Labs' compact, highly-integrated RF solutions and expertise in high-performance, mixed-signal MCUs enable customers to quickly develop reliable, cost effective RKE and remote control solutions. This combined RF and MCU expertise provides a one-stop solution for all your wireless needs.



EZRadio

Part Number	Type	Modulation Scheme (Max. Data Rate)	Frequency Bandwidth (MHz)	Frequency Range (MHz)	Output Power (Max. dBSm)	Supply Voltage	Wake-Up Receiver	TRF	Free Space Tx Current (mA)	Start-Up Time (ms)	Idle Current (mA)	Package		
S14020	TX	256	Yes	Yes	1	3	2.2-5.4V	—	25-75	250 µs	1.3 mA	TSSOP16		
S14021	TX	115	512	—	Yes	6	8	2.2-5.4V	—	1-bit	25-75	250 µs	1.5 mA	TSSOP16
S14022	TX	115	—	Yes	6	8	2.2-5.4V	—	64-bit	25	300 µs	0.5 mA	TSSOP16	
S14200	RX	256	1	Yes	Yes	—	2.2-5.4V	—	10-bit	2.5-7.5	250 µs	3.0 mA	TSSOP16	
S14322	RX	256	—	Yes	Yes	—	2.2-5.4V	—	64-bit	25	300 µs	0.5 mA	TSSOP16	
S14420	TRX	256	—	Yes	Yes	2	3	2.2-5.4V	—	10-bit	2.5-7.5	250 µs	3.0 mA	TSSOP16
S14421	TRX	115	—	Yes	Yes	7	5	2.2-5.4V	—	10-bit	2.5-7.5	200 µs	0.6 mA	TSSOP16

EZRadioPRO

Part Number	Type	Modulation Scheme (Max. Data Rate)	Frequency Range	Output Power (Max. dBSm)	Sensitivity (dBm)	TX Current (mA)	Package
S14030	TX	128 kbps	40 kbps	900-960 MHz	-10 to +13	—	QFN16
S14031	TX	128 kbps	40 kbps	240-930 MHz	-10 to +13	—	QFN16
S14032	TX	128 kbps	40 kbps	240-930 MHz	+11 to +20	—	QFN16
S14033	RX	128 kbps	40 kbps	900-960 MHz	-118 dBm	-119 dBm	QFN20
S14034	RX	128 kbps	40 kbps	240-930 MHz	-118 dBm	-119 dBm	QFN20
S14035	TRX	128 kbps	40 kbps	900-960 MHz	-118 dBm	-119 dBm	QFN20
S14036	TRX	128 kbps	40 kbps	240-930 MHz	-118 dBm	-119 dBm	QFN20

FEATURES

EZRadioPRO

- 240-960 MHz continuous frequency coverage
- Configurable output power up to +20 dB
- Ultra low power shutdown mode : 10 nA
- GFSK, FSK, OOK modulations
- Automatic PA ramp-up/down to prevent spectral splatter
- Fast frequency-hopping capability
- Configurable data packet handler
- Embedded antenna diversity algorithm
- Integrated low battery detector, temperature sensor, power-on-reset, general purpose ADC, and 32.768 KHz RC or XTAL
- Integrated 64-byte transmitter and receiver FIFOs

EZRadio

- 315, 434, 868 and 915 MHz band support
- Fully integrated (low BOM, easy design-in)
- Stable and accurate FSK modulation with programmable deviation
- Fast-settling, programmable, high-resolution PLL synthesizer
- Automatic antenna tuning
- Fast frequency-hopping capability
- Direct loop/IFA antenna support
- No alignment required in production
- SPI interface

APPLICATIONS

- Remote keyless entry (RKE)
- Aftermarket car alarms
- Remote controls, IR replacement/extension
- Satellite set-top boxes
- Wireless weather stations
- Home security and automation
- Garage and gate door openers



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